



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination, May-2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Biomedical) / B.Tech. (Biotech) / B.Tech. (Chemical) / B.Tech. (Civill) / B.Tech. (Mechanical) / B.Tech. (CSE) / B.Tech. (CSE-AIML) / B.Tech. (CSE-Cyber Security) /B.Tech. (Foodtech) / B.Tech. (Electrical)
Semester/Year:	II Sem.
Paper Code:	BTESSDS202T22
Title of Paper:	Smart and Sustainable Energy
Time: 3 hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

Attempt any five Questions from section A and attempt any four Questions from section B.

Question Number	SECTION-A	Marks Allotted
Q1.	Justify the current status of various modes of renewable power generations in India.	4
Q2.	Discuss the parameters of an Energy Storage System.	4
Q3.	Explain the Policy frameworks for better integration of energy resources and technology.	4
Q4.	How do photovoltaic cells in solar panel harness sunlight to generate electricity?	4
Q5.	Define and Explain the functions of Smart Grid.	4
Q6.	Compare conventional and non-conventional energy resources.	4
Q7.	What are the main categories of renewable energy sources, and how do they differ in terms of their generation	4

mechanisms and environmental impacts?

SECTION-B

- Q1.** Interpret World Energy Scenario with respect to consumption of fossil fuels using relevant statistics. **10**
- Q2.** Define the following terms: **10**
- a.) Green Buildings
 - b.) Sustainability
 - c.) Energy Efficiency Technologies
 - d.) Intermittency & Variability
 - e.) CHP
- Q3.** Explain Major Challenges and opportunities of Renewable Energy. **10**
- Q4.** Explain the importance of Energy Storage System. **10**
- Q5.** Describe the operation of a hydroelectric power plant using a labelled diagram. **10**
- Q6.** What is an Electrical Grid? Explain Its Infrastructure with the help of diagram. **10**